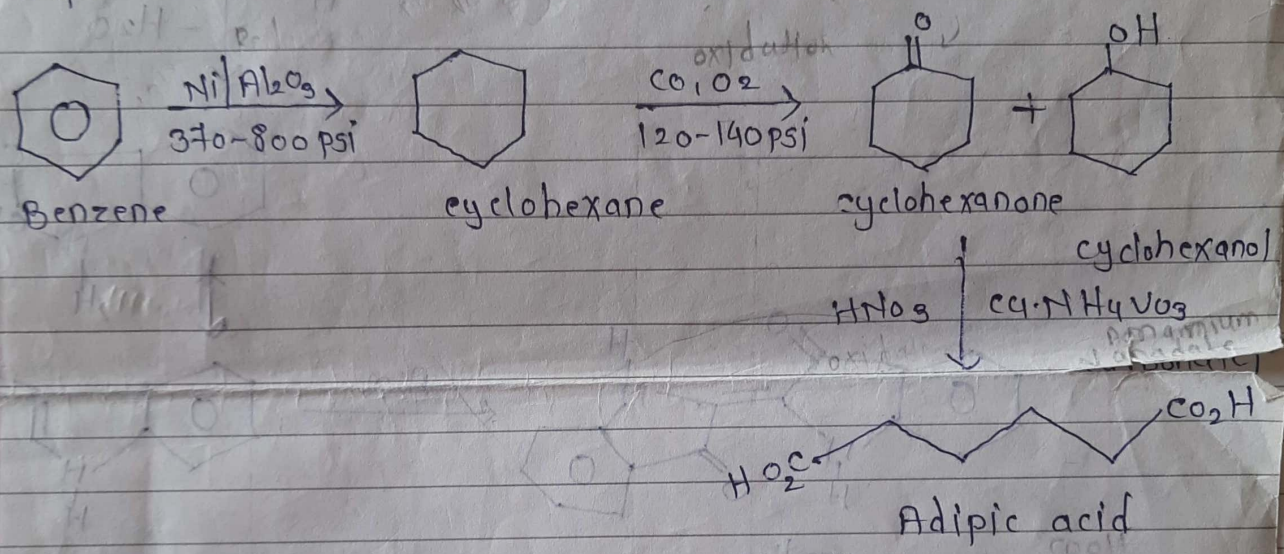
Explain the conventional and green synthesis of Adipic acid. Mention the green chemistry principle involved.

Synthesis of Adipic acid

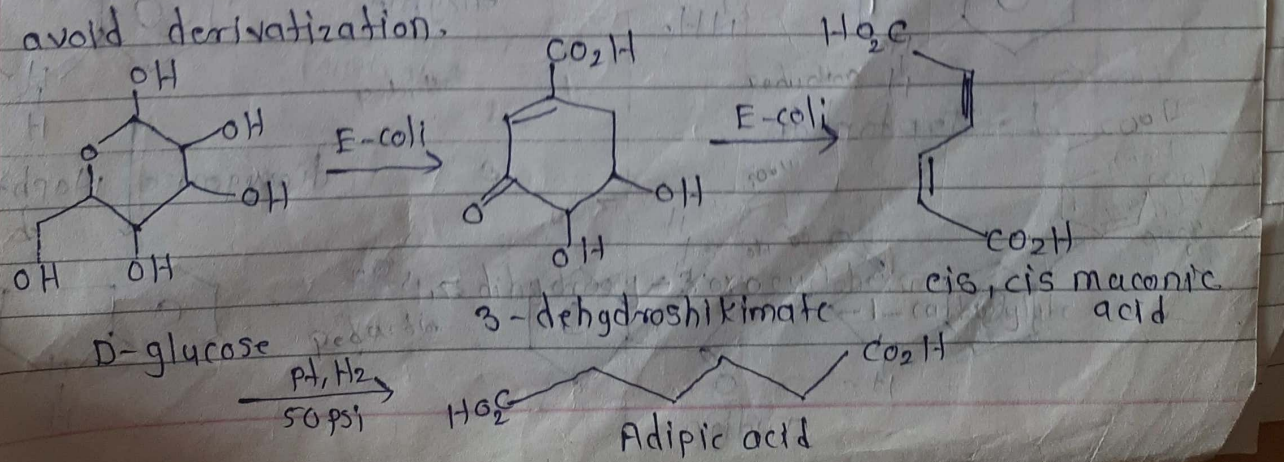
→ conventional route pathway

- For synthesis of adipic acid, molecule, starting material benzene is treated with Ni/Alumina catalyst at high temp. to get saturated cyclic hydrocarbon and cyclohexane.
- This is treated with carbon monoxide in presence of oxygen at medium pressure to get mixture of cyclohexanone and cyclohexanol as intermediate which are converted into adipic acid.



Green route pathway

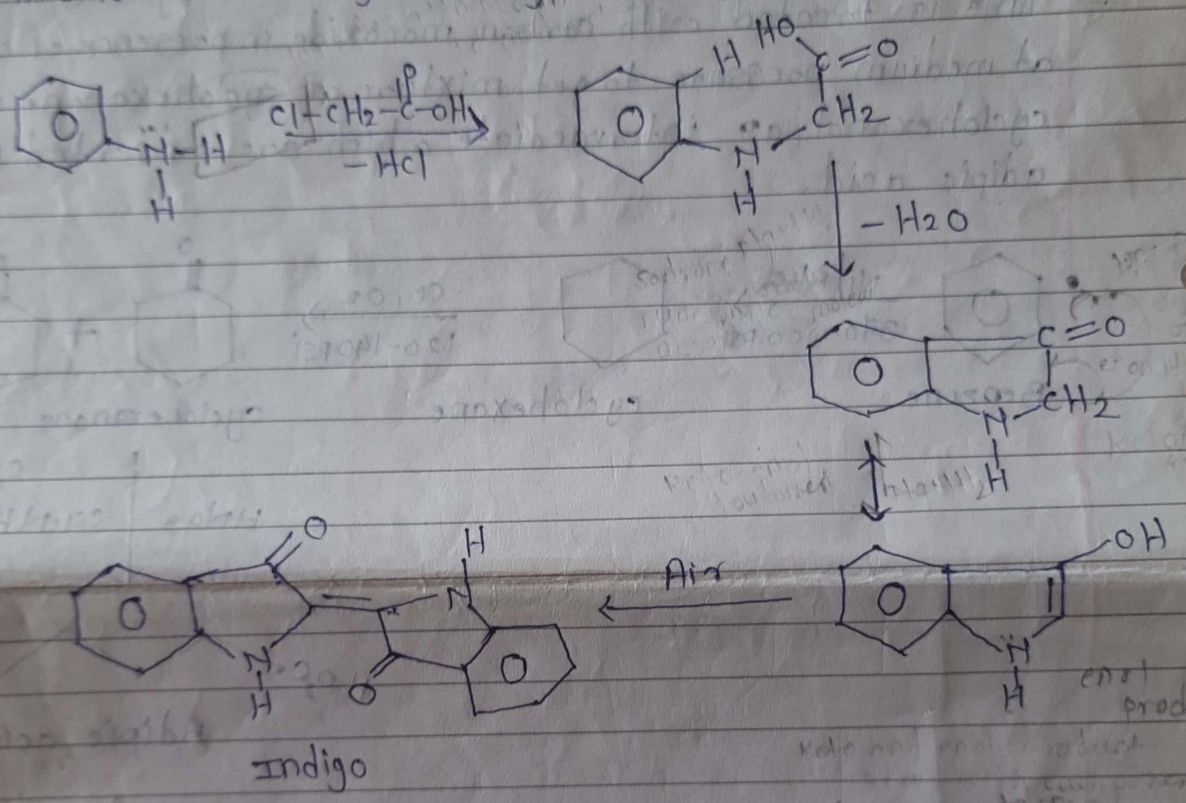
- Glucose treated with E. coli to get 3-dehydroshikimate as first intermediate, cis, cis muconic acid as second intermediate. Further it is react with Pt. hydrogen at 50 psi to give adipic acid.
- This is a example of avoid hazardous chemicals and avoid derivatization.



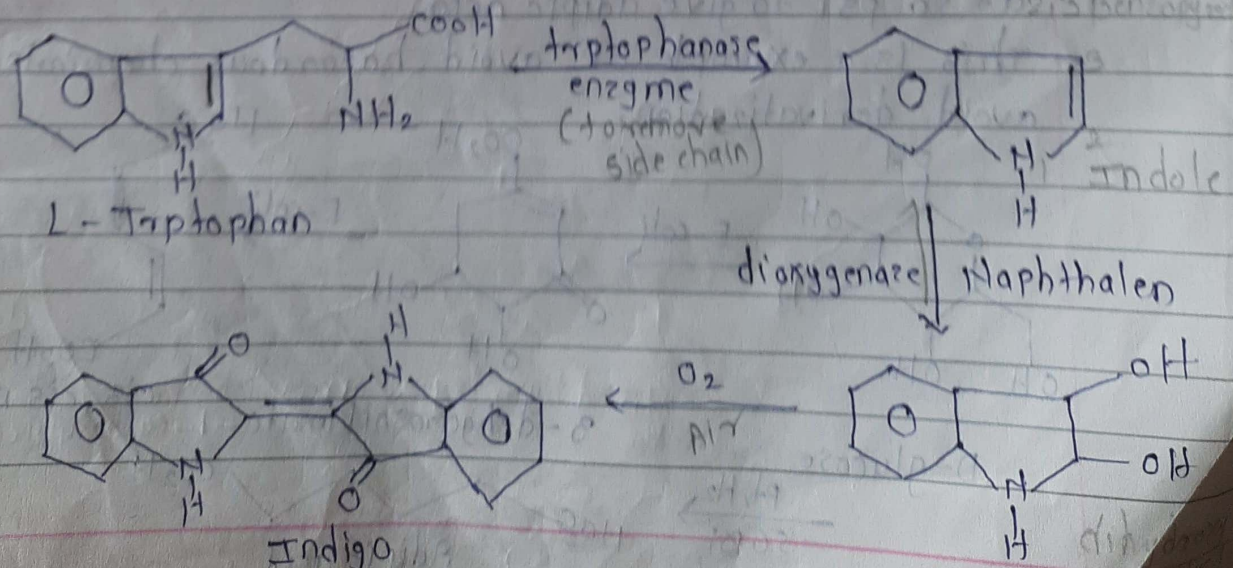
Explain the conventional and green route of manufacturing of Indigo dye. Also mention principle involved
- This is example of Avoid use of hazardous chemicals

Synthesis of Indigo

- Conventional route pathway
- Aniline is treated with chloroacetic acid and the resultant heterocyclic carboxylic acid which is treated with sodalime to get a heterocyclic phenol
- This is followed by coupling reaction in air forming Indigo dye



- Greener route pathway
Reaction in which ~~L-tryptophan~~ tryptophan is removed enzymatically to give Indole. It can be dehydroxylated enzymatically and then oxidized with oxygen to Indigo

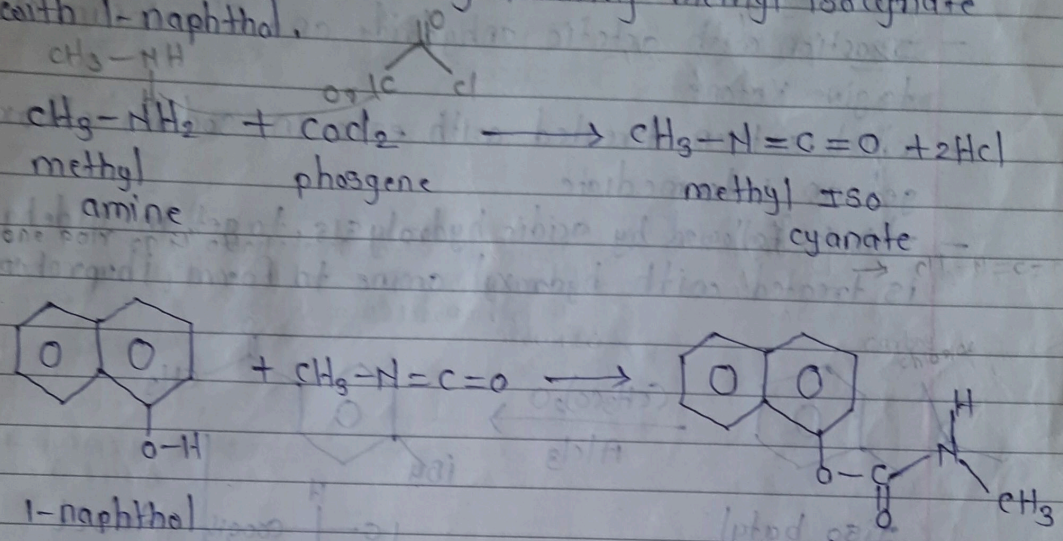


Explain the conventional and green synthesis of carbaryl. Mention principle involved in it.

- This is example of safer chemistry for accident prevention

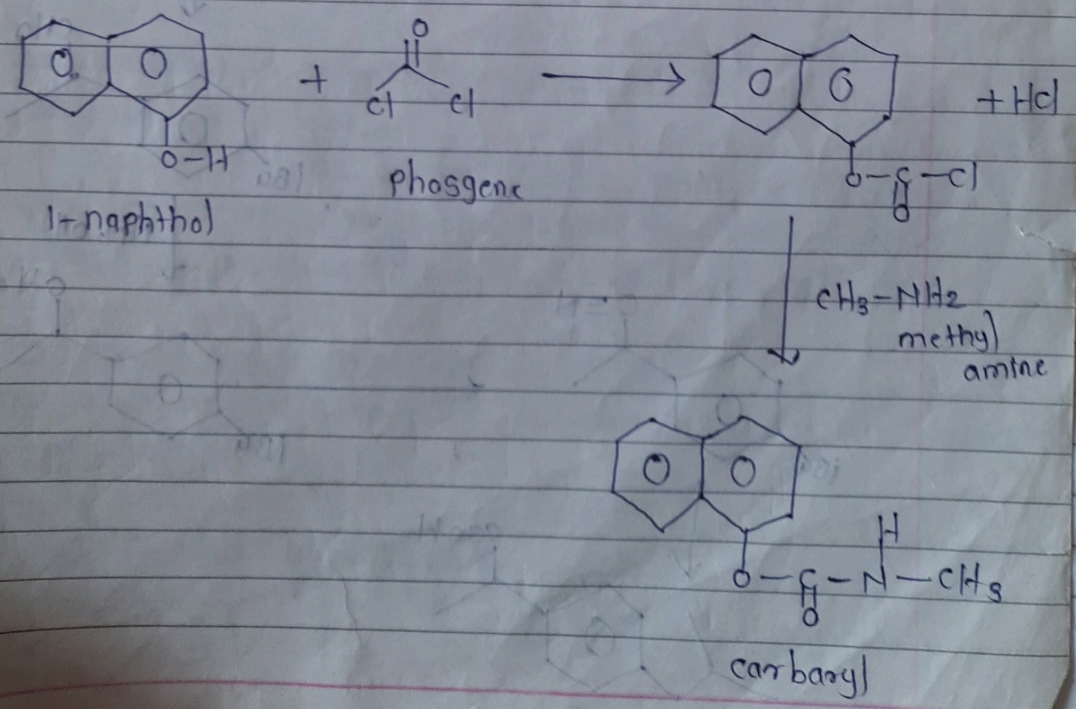
c Synthesis of carbaryl

- conventional route pathway
- Amine is treated with phosgene to get methyl isocyanate
- carbaryl is prepared by treating methyl isocyanate with 1-naphthol.



- Greener route pathway

- 1-napthol react with phosgene which is then ~~conve~~ treated with methylamine



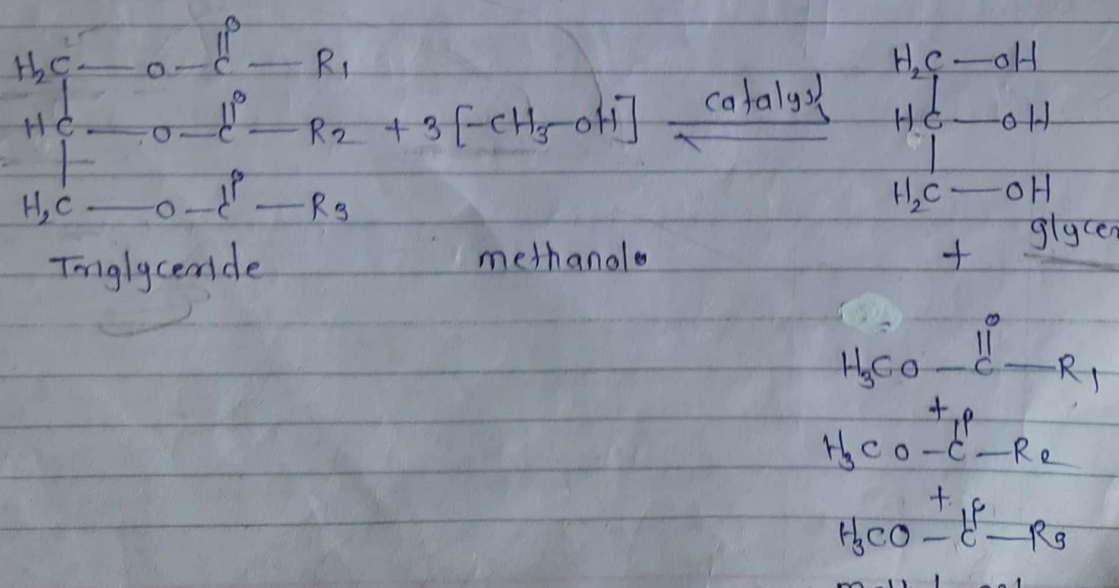
What is biodiesel? Explain the trans-esterification method for the synthesis of biodesel? Mention the advantages of biodiesel as fuel.

Green fuel - Biodiesel

- chemically biodiesel is the methyl esters of long chain carboxylic acids
- Biodiesel is obtained by transesterification of vegetabe oil or animal fats with methyl alcohol using sodium metal or sodium methoxide as catalyst.
- Transerification is the process of converting one ester to another ester.

Biodiesel synthesis

- Filter the vegetable oil / fat
- Heat it at 110°c with stirring to remove any water from it
- Sodium methoxide and methanol is then added to it
- The mixture is then heated for half an hour with stirring
- The reaction mixture is then cooled and mixed with sufficient amount of water
- The glycerol and soap gets dissolved in the water phase
- water insoluble phase (Biodiesel) is separated ~~and~~.
- antioxidant added to the biodiesel to avoid oxidation and polymerization.
- The product is given the name biodiesel as it is obtained from biological product and it is biodegradable material or green fuel



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Advantages of Biodiesel

1. It is prepared form renewable resources
2. It is biodegradable and non-toxic
3. The exhaust gas emission is lesser as compared to the conventional diesel fuel
4. It is free from sulphur and aromatics
5. It is safer to handle and easy to manufacture
6. Like any other fuel, it can also be used as a heating fuel in domestic and commercial boilers

Explain synthesis of Ethanol?

Green fuel - Ethanol

- Ethanol is prepared by the process of Fermentation
- Fermentation is a biochemical process which occurs in the presence of enzymes secreted by micro-organism such as yeast.

Ethanol synthesis

Sucrose in molasses is converted into glucose/fructose by Invertase enzyme. Finally zymase enzyme converts glucose/fructose to desired ethanol product.

$$\underset{\text{sucrose in molasses}}{C_{12}H_{22}O_{11}} \xrightarrow[H_2O]{\text{invertase}} \underset{\text{glucose}}{C_6H_{12}O_6} + \underset{\text{fructose}}{C_6H_{12}O_6}$$

$$\underset{\text{glucose / Fructose}}{C_6H_{12}O_6} \xrightarrow{\text{zymase}} \underset{\substack{\text{ethanol} \\ \text{(ethyl alcohol)}}}{2C_2H_5OH} + 2CO_2$$

when the ethanol is mixed with petrol and used as fuel in internal combustion engine, it is known as Power Alcohol. Genrally, 2-25% ethanol is mixed with petrol

Advantages

- Ethyl alcohol has good antiknocking property and its octane no. is 90, while octane no. of petrol is 65 Therefore addition of ethanol to petrol increases its octane no.
- Alcohol has property of absorbing any traces of water if present in petrol
- Power alcohol is cheaper than petrol
-

6.4.4 Ionic Liquid

Q. Define ionic liquid. Mention the characteristics of ionic liquid **(3 Marks)**

- **Ionic liquid** : Ionic liquids are conventionally defined as compounds completely composed of ions with melting point below 100 °C.
- **Examples :** Typical combination of cations and anions in ionic liquids are mentioned below;

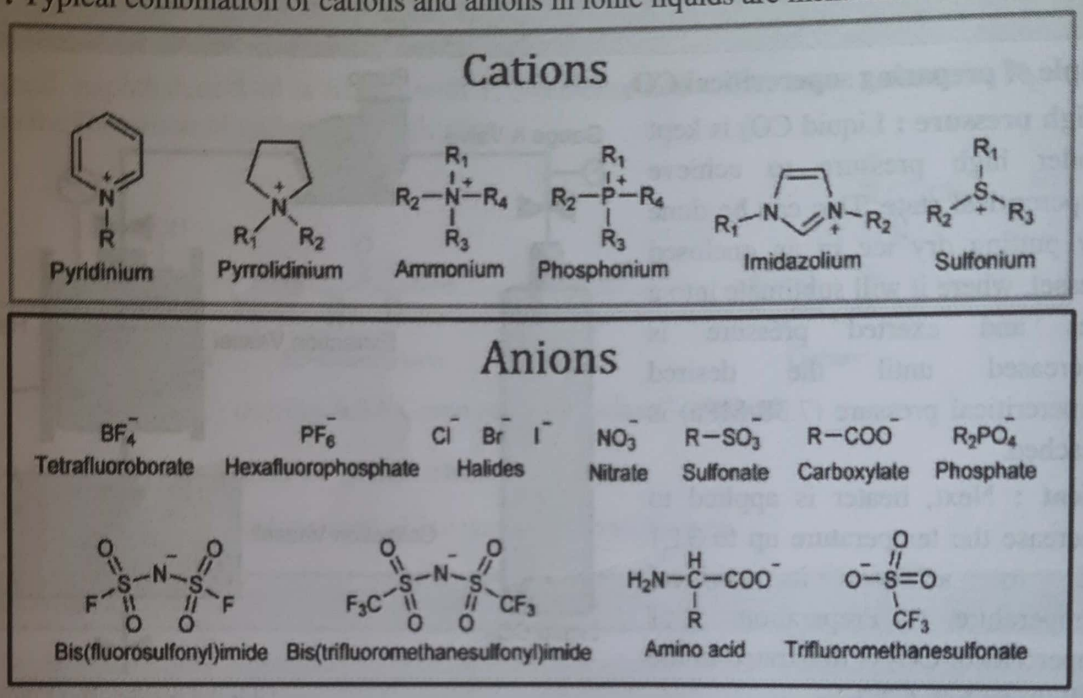

Fig. 6.4.3

6.4.6 Characteristics of Ionic Liquids

1. Ionic liquids have low volatility.
2. Ionic liquid are highly resistive towards the changes of its physical and chemical state even at higher temperature.
3. These have low vapor pressure.
4. Ionic liquids have low combustibility.
5. Ionic liquids are generally hydrophilic in nature but it depends upon nature of cation and anions. If the ion counterpart contains long alkyl chain, it makes ionic liquid hydrophobic.
6. These can be used in a wide electrochemical potential window without being oxidized or reduced.
7. More importantly, ionic liquids have intrinsic ion conductivity.

6.4 GREEN SOLVENTS

Q. What is green solvent? **(2 Marks)**

Green Solvent : Green solvents are the biosolvents which are environment friendly, less toxic, recyclable and bio-degradable. E.g. Water, γ-Valerolactone, Supercritical CO_2, Glycerol etc.

Q. Define the term supercritical solvent. Give the characteristics of supercritical solvents. **(5 Marks)**

Supercritical solvent

Supercritical solvents (SCFs) are the compressed fluids that carries the properties of both gas and liquid.

E.g. Most commonly used supercritical solvent is supercritical CO_2. It has critical temperature of 31.1°C and a critical pressure of 7.38 MPa.

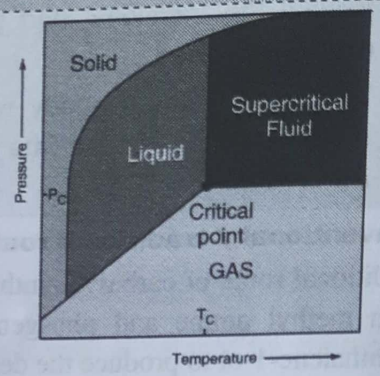

Fig. 6.4.1 : P-T Diagram of supercritical solvent

6.4.1 Characteristics Supercritical Solvents (SCFs)

1. **Gas-like behavior**: SCFs are compressible fluid that can fill the container where they are placed.
2. **Liquid-like behavior**: SCFs have almost similar densities and solvating power similar to liquids.
3. **Variable properties**: In comparison to regular liquid, SCFs can have different properties and their densities and viscosity can change with temperature and pressure.
4. **Phase**: SCFs don't posses definite phase, and it's impossible to liquefy matter by changing the pressure.
5. **Diffusivity**: SCFs have high diffusivity, which allows them to permeate solids and improve mass transfer.
6. **Viscosity**: SCFs have low viscosity as it is observed for gas.
7. **Surface tension**: SCFs don't have any surface tension (surface tension $\gamma = 0$).